

Computational Astrophysics Exercises — Assignments Set 3

1. Catastrophic cancellation

Consider the quadratic equation

$$ax^2 + bx + c = 0, \quad (1)$$

with the solutions

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \quad (2)$$

Apparently, there is a problem with subtractive cancellation for x_1 if $b^2 \gg 4ac$, when the nominator becomes $-b + \sqrt{b^2 - 4ac} \approx -b + \sqrt{b^2} = -b + b$.

- Write a program that calculates all solutions for arbitrary values of a , b , and c .
- Investigate how errors in your computed answers become large as the subtractive cancellation increases and relate this to the known machine precision. (Hint: A good test case is $a = b = 1$, $c = 10^{-n}$, $n = 1, 2, 3, \dots$)
- Implement an alternative way to calculate x_2 , where subtractive cancellation is avoided for $b^2 \gg 4ac$.
- Modify your program from a) to automatically choose the implementation that gives the most accurate result.

(Ob011 points)

2. Interpolation¹

Consider the Runge function

$$f(x) = \frac{1}{1 + x^2}. \quad (3)$$

Approximate the Runge function by a polynomial of grade n using nodes in the interval $[-5, 5]$.

- Choose the equidistant $(n + 1)$ nodes

$$x_k = -5 + \frac{10}{n}k \quad \text{where} \quad k = 0, \dots, n. \quad (4)$$

Calculate and plot the interpolation polynomial $P_n(x)$ for $n = 12$ and $n = 20$. Compare with the analytical function $f(x)$. Note: Use Newton's divided differences interpolation polynomial as shown in the lecture to calculate the coefficients c_i of the polynomial $P_n(x)$

¹For comparison and playing around: fitting tools of numpy are given by the functions `numpy.polyfit` and `numpy.polyval`

<http://docs.scipy.org/doc/numpy/reference/generated/numpy.polyfit.html>.

```

1 for i = 0, 1, ..., n
2   c_i = y_i
3 for k = 1, 2, ..., n
4   for i = n, n - 1, ..., k
5     c_i = (c_i - c_{i-1}) / (x_i - x_{i-k}).

```

In order to compare the interpolated with the analytical values, you need to calculate the polynomial $P_n(x)$ at further abscissae x_v . This can be done using the Horner-Scheme, see the lecture slides on ILIAS.

```

1 p = c_n
2 for k = n-1, n-2, ..., 0
3   p = c_k + (x_v - x_k) p.

```

b) Calculate and plot the value of the error function

$$W(x) \equiv \prod_{i=0}^n (x - x_i) \quad (5)$$

c) Repeat the complete analysis for the non-equidistant nodes

$$x_k = \frac{b-a}{2} \cos\left(\frac{n-k}{n}\pi\right) + \frac{b+a}{2} \quad \text{for } k = 0, \dots, n, \quad (6)$$

where the limits of the interval are given by $a = -5$ and $b = 5$. What do you find?

(ob111 points)

in total

(ob1010 points)

Deadline: Friday, 15 May 2026 at noon 🍏, via email (including code) to christoph.schaefer@uni-tuebingen.de.